FINAL REPORT INDUSTRIAL TRAINING

AT MAPSKART TECHNOLOGY SDN BHD

PROJECT TITLE:

Data-Driven Analysis of Tree Suitability for Urban Roadsides

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ABSTRACT

This industrial training report documents the practical training undertaken by a student of Universiti Malaysia Terengganu (UMT) at Mapskart Technology Sdn. Bhd. for a one-year period from 19 March 2025 to 19 March 2026, as part of the Bachelor of Science in Data Analytics programme. The training aimed to provide practical exposure to data analytics and real-world problem solving within a professional working environment. The primary focus of the industrial training was a data-driven project on urban tree suitability assessment in Malaysia. Urban planners and organisations often face challenges in selecting suitable trees for roadside environments due to variations in environmental, structural, and health-related factors. To address this issue, an analytical study was conducted using urban tree data obtained from the Greehill urban data platform, focusing on the Seberang Perai area in Penang, Malaysia. The project involved data cleaning, preprocessing, validation, and analysis using Microsoft Excel, Python, and Power BI. A Multi-Criteria Decision Analysis (MCDA) approach was applied to evaluate key indicators influencing tree suitability, while selected supervised machine learning models were explored to support suitability classification within a decision-support framework. The analytical results were presented through dashboards and an intelligent system to support data-driven planning and operational decisions. Through this industrial training, the student developed strong technical competencies in data analytics as well as professional skills such as communication, teamwork, and problem solving in an industry-based environment.

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CHAPTER 1 : INTRODUCTION

1.1 Project Background

Urban forestry plays an important role in enhancing environmental sustainability, public safety, and overall urban livability. However, inappropriate tree selection or placement may pose risks to road users, surrounding infrastructure, and long-term maintenance operations. As urban areas continue to expand, organisations involved in urban planning and environmental management are increasingly required to make informed and evidence-based decisions regarding tree suitability and placement.

Mapskart Technology Sdn. Bhd. is a company that provides data-driven and geospatial solutions, including customised software systems and Internet of Things (IoT) applications related to environmental and urban analysis. In line with this, the industrial project focuses on supporting urban tree management through analytical methods. The project explores how data analytics techniques can be applied to assess roadside tree suitability by considering environmental benefits, structural conditions, and operational safety requirements.

1.2 Problem Statement

At the initial stage of the industrial training, the company did not have a dedicated data analytics team, and the existing urban cloud platform hosted by Greehill, a partner of Mapskart Technology Sdn. Bhd., did not provide integrated tree suitability assessment or structured risk evaluation for urban roadside trees in Malaysia. As a result, information related to tree suitability and potential risks was not readily available in an analytical and decision-support format.

In general practice, urban roadside tree assessment is often based on fragmented datasets, manual inspection, or subjective judgement. Such approaches may lead to

inconsistent decision-making, difficulties in maintenance prioritisation, and challenges in identifying suitable trees for specific urban environments. The absence of a structured, data-driven analysis limits the organisation's ability to systematically evaluate tree condition, environmental factors, and structural risks. Consequently, both the company and its stakeholders face challenges in supporting planning and operational decisions using quantitative evidence. This situation highlights the need for a reliable analytical framework that can process existing data and generate meaningful insights for urban tree suitability assessment.

1.3 Project Objectives

The objectives of this industrial project are as follows:

1. To analyse existing urban tree datasets and determine the suitability of trees planted at specific roadside locations.
2. To evaluate the relative importance of environmental, health, and structural indicators in assessing urban tree suitability using a multi-criteria approach.
3. To explore supervised machine learning models based on historical data to support tree suitability classification.
4. To develop a tree suitability intelligent system that supports informed decision-making for both the company and its clients.
5. To present analytical results in a clear and interpretable manner for technical and non-technical stakeholders.

1.4 Project Scope and Constraints

This project focuses on the analytical evaluation of existing urban tree datasets provided by the company, with emphasis on environmental performance, tree condition, and roadside suitability. The scope of the project is limited to data-driven analysis and system development, and does not include physical inspection, on-site validation, or field measurements of trees.

Several constraints were identified during the project, including limited time for iterative model refinement, dependency on the availability and structure of client-provided datasets, and restricted access to certain proprietary information. These constraints were considered throughout the project to ensure that the proposed analytical framework and system development remained realistic, ethical, and aligned with industry practices.

1.5 Significance of the Project

This project contributes to the organisation by demonstrating how data analytics can be applied to support structured and objective assessment of urban tree suitability. The analytical outcomes provide quantitative insights that may assist decision-makers in maintenance prioritisation, risk awareness, and long-term urban environmental planning.

From an academic perspective, the project enables the application of data analytics concepts in a real-world industrial environment, thereby bridging theoretical knowledge with practical implementation. The findings highlight the potential of data-driven approaches in addressing operational challenges within the urban environmental management domain.

CHAPTER 2 : TECHNICAL CONTEXT AND METHOD JUSTIFICATION

2.1 Overview of the Analytical Approach

This industrial project adopts a data-driven analytical approach to assess the suitability of trees planted along urban roadsides. The approach focuses on processing existing tree datasets to extract meaningful insights that support planning and decision-making. Rather than relying on manual assessment or subjective judgement, structured data analytics techniques are applied to evaluate multiple factors influencing tree suitability.

The analytical workflow includes data cleaning, preprocessing, exploratory data analysis (EDA), suitability scoring using Multi-Criteria Decision Analysis (MCDA), and predictive modelling using supervised machine learning models. MCDA is a structured method used to evaluate alternatives based on multiple criteria simultaneously. In this project, it is applied to score each tree using environmental, health, stability, and risk-related indicators, allowing trees to be compared and prioritised systematically.

These steps enable the organisation to assess tree conditions in a consistent and interpretable manner while ensuring that the analysis remains aligned with real-world operational requirements.

2.2 Data Analytics and Modelling Techniques

Descriptive analytics was first applied to understand data distributions, identify patterns, and explore relationships between variables such as environmental conditions, tree health, and structural stability. This helped identify which factors affect tree suitability and guided the selection of features for predictive modelling.

For predictive analysis, supervised machine learning models including Random Forest, XGBoost, and Multi-Layer Perceptron (MLP) were considered using historical tree data:

- Random Forest: A group of decision trees that work together to make predictions. It is easy to understand and works well with different types of data.
- XGBoost: An advanced method that builds many small decision trees step by step, learning from previous mistakes to make more accurate predictions.
- MLP (Multi-Layer Perceptron): A type of artificial neural network that can learn complex relationships between input features and output predictions.

These models were considered because they are capable of handling complex patterns in structured data, are suitable for classification tasks, and are commonly applied in predictive analytics. The final selection of the most appropriate model is based on comparative performance evaluation and is presented in Chapter 4: Results and Discussion. The modelling process involves data preprocessing, feature selection, model training, and basic validation to ensure reliable performance. The outputs from these models will later be used to support decision-making through the Tree Suitability Intelligent System.

2.3 Tool and Technology Selection

The tools and technologies were selected based on suitability for data analytics tasks, industry acceptance, and ease of integration. Python served as the primary programming language due to its extensive support for data analysis and machine learning libraries, including pandas, NumPy, scikit-learn, and matplotlib. Microsoft Excel was used for preliminary data inspection and cleaning, while Power BI was used to visualise results and present insights interactively. Visual Studio Code served as the main development environment for coding and experimentation. The choice of these tools ensures that the analytical workflow is practical, reproducible, and aligned with common industry practices.

2.4 Industry Practices and Method Justification

The analytical methods adopted reflect standard practices in industry-based data analytics and data science projects. Structured data preprocessing, exploratory analysis, and model development are standard steps in transforming raw data into actionable insights. Using interpretable models and visualizations ensures transparency, allowing stakeholders to understand the reasoning behind outputs rather than relying on fully opaque models.

The combination of suitability scoring and predictive modelling was chosen to support operational decision-making, such as identifying high-risk trees, prioritising maintenance, and guiding future planting strategies. This approach delivers both immediate insights and forward-looking support for the organisation.

2.5 Integration of Theory and Practice

From an academic perspective, this project applies concepts such as data preprocessing, feature analysis, and supervised learning within a real industrial context. Theoretical knowledge gained during coursework was adapted to accommodate real-world data limitations, time constraints, and organisational requirements.

The development of the Tree Suitability Intelligent System demonstrates the practical application of analytics, providing fast and interpretable insights to both technical and non-technical stakeholders. This integration highlights the importance of balancing technical rigor with usability when applying analytics in an industrial environment.

CHAPTER 3 : METHODOLOGY

3.1 Data Collection

The data used in this project was obtained from the Greehill Urban Tree Inventory platform and collected by Mapskart Technology Sdn. Bhd. using car-mounted LiDAR sensors and high-resolution cameras.

Figure 3.1.1 shows car-mounted LiDAR sensors and high-resolution cameras

These technologies were used to record streetside trees along urban roads, producing detailed digital information for each tree. Using deep learning techniques, each tree was converted into a digital representation and assessed remotely by certified arborists through the Greehill platform.

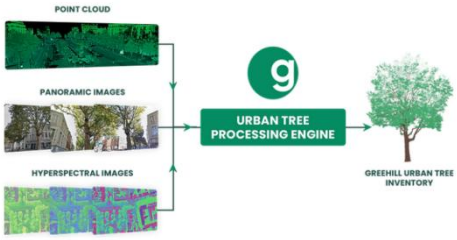

Figure 3.1.2 shows 4D digital tree twins

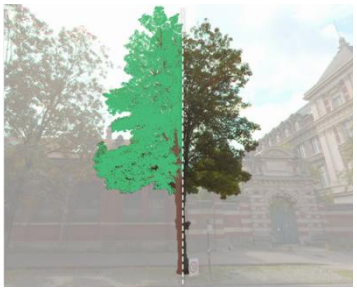

Figure 3.1.3 shows tree image represent in point cloud format

For every tree, attributes such as location, physical characteristics, health condition, environmental contribution, and risk indicators were recorded and stored in the Greehill Urban Cloud.

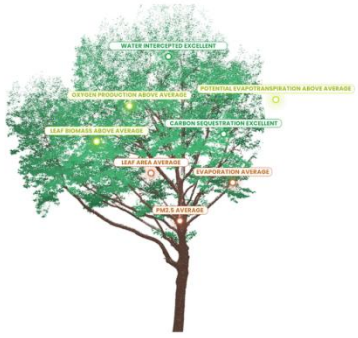

Figure 3.1.4 shows attribute details of the tree.

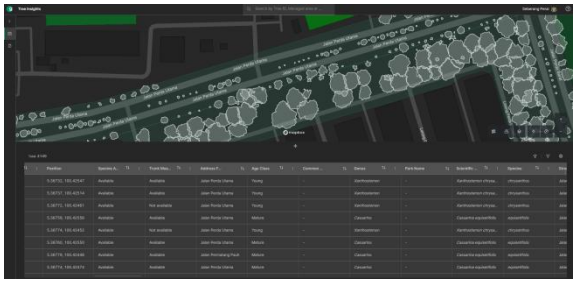

Figure 3.1.5 shows attributes stored in Greehill Urban Cloud platform

The dataset was downloaded in CSV format and focuses specifically on Seberang Perai, Penang, Malaysia. It represents urban roadside trees collected for operational monitoring and environmental management purposes. The final dataset consists of 4,149 tree records and 61 attributes, including:

- Tree identification details (*Tree ID, genus, species*)
- Location information (*street name, position, coordinates*)
- Physical characteristics (*height, diameter at breast height*)
- Condition indicators (*condition, status*)
- Environmental and risk-related attributes (*oxygen production, dieback, slenderness*)

This dataset serves as the primary data source for all analysis conducted in this project.

3.2 Data Cleaning

Data cleaning was performed to improve data quality and ensure consistency before analysis. Microsoft Excel was used for the initial cleaning process due to its efficiency in handling large tabular datasets and allowing quick inspection of missing values.

The following steps were applied:

- Empty or non-informative columns were removed.

- Missing categorical values were replaced with “Unknown”.
- Missing numerical values were filled using the median of each column.
- Duplicate records were checked and verified.

Approximately 1,900 missing values were identified, representing about 47% of the dataset. Instead of removing these records, missing values were imputed to prevent information loss and reduce potential bias. Categorical missing values were replaced with “Unknown” to preserve the original data distribution and acknowledge the absence of information rather than making assumptions. Numerical missing values were filled using the median because the data distribution was skewed and contained outliers, making the median more robust than the mean.

After cleaning in Excel, the dataset was validated using Python to confirm that no missing values or inconsistencies remained. These steps ensured that the dataset was complete and suitable for exploratory analysis, MCDA, and machine learning without removing meaningful records.

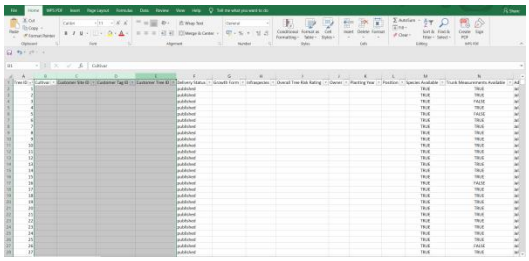

Figure 3.2.1 shows empty columns were removed.

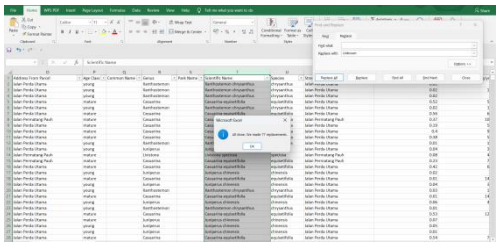

Figure 3.2.2 shows missing values filled in.

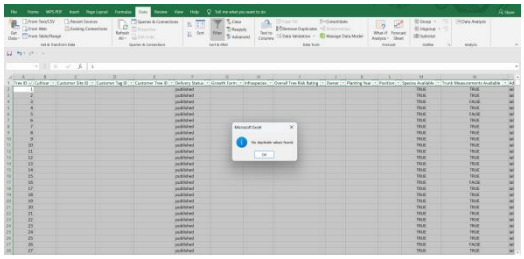

Figure 3.2.3 shows there are no duplicates

3.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the structure, distribution, and relationships within the dataset before applying MCDA and machine learning models. This step aimed to identify general patterns, data distributions, and potential anomalies that could influence subsequent analysis.

EDA was performed using Python to ensure reproducibility and analytical flexibility. The following techniques were applied:

- Bar charts to analyse categorical variables such as genus, condition, and vitality.
- Histograms to examine the distribution of numerical variables, identify skewness, and observe the presence of extreme values.

The insights obtained from EDA informed decisions related to data transformation, feature selection, and weighting strategies in the MCDA framework. This ensured that subsequent analyses were grounded in an initial understanding of the dataset characteristics.

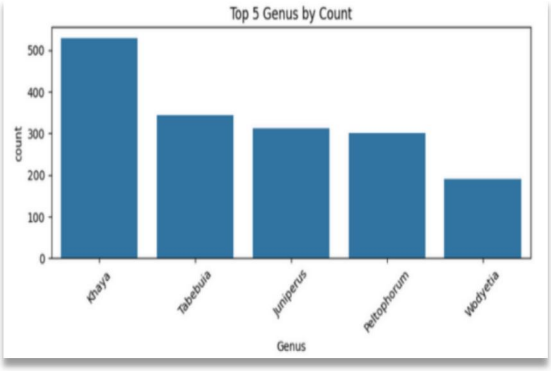

Figure 3.3.1 shows a bar chart showing the top five most common Genus.

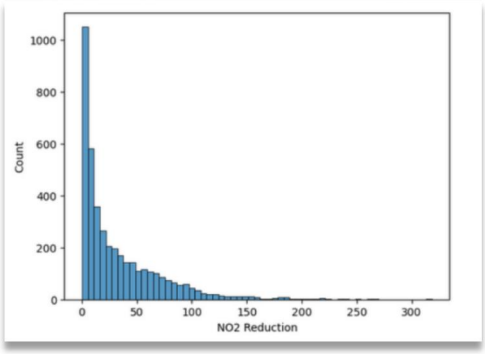

Figure 3.3.2 shows a histogram representing the distribution of NO2 Reduction.

3.4 Data Transformation

3.4.1 Outlier Analysis

Outlier treatment was conducted to reduce the influence of extreme values that could distort the analysis and affect the performance of MCDA and machine learning

models. Boxplots were used to visually identify outliers in numerical variables. The Interquartile Range (IQR) method was applied to detect outliers. The IQR is calculated as shown in Equation (1):

$$IQR = Q3 - Q1$$

Data points falling below the lower bound ($Q1 - 1.5 \times IQR$) or above the upper bound ($Q3 + 1.5 \times IQR$) were classified as outliers. These outlier values were removed from the dataset to minimise noise and prevent extreme measurements from disproportionately influencing the scoring process and predictive models. This approach helps improve the robustness and reliability of subsequent analyses, particularly when applying normalization, MCDA weighting, and machine learning algorithms that are sensitive to extreme values.

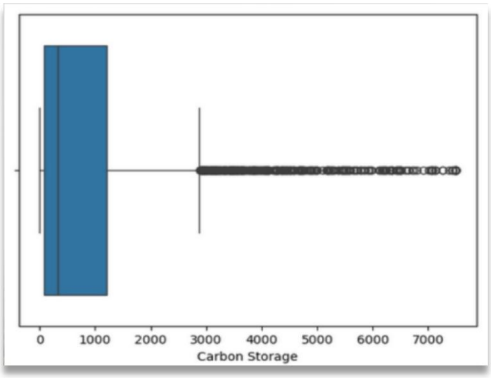

Figure 3.4.1.1 shows boxplot of carbon storage showing presence of outliers.

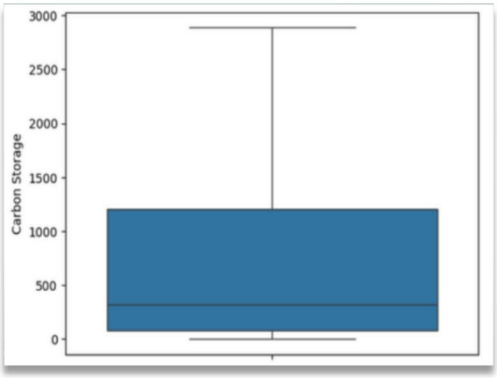

Figure 3.4.1.2 shows boxplot after outlier analysis.

3.4.2 Categorical Encoding

Categorical variables were transformed into numerical representations to enable machine learning analysis. Two encoding techniques were applied:

- Label Encoding was used for structured categorical variables such as Genus, Species, Scientific Name, Street Name and Number, and Managed Areas, where unique numerical identifiers were sufficient.

- One-Hot Encoding was applied to nominal categorical variables such as Age Class, Condition, and Vitality. This method converts categories into binary indicator variables and prevents the model from assuming an ordinal relationship between categories.

3.4.3 Data Normalization

Numerical features were normalized to ensure all variables were on a comparable scale and to prevent features with larger magnitudes from dominating the analysis. Min–Max scaling was applied to transform values into a range between 0 and 1.
The Min–Max normalization is defined as in Equation (2):

$$X_{normalized} = X - X_{min} / X_{max} - X_{min}$$

where:

- X = Feature value to be normalized
- $Xmin$ = Minimum feature value in the dataset
- $Xmax$ = Maximum feature value in the dataset

3.5 Multi-Criteria Decision Analysis (MCDA)

Multi-Criteria Decision Analysis (MCDA) was used to assess the suitability of each tree based on multiple criteria simultaneously. This approach allows complex tree characteristics to be evaluated in a structured and transparent manner, rather than relying on a single indicator or subjective judgement.

Prior to MCDA computation, categorical variables that were transformed using one-hot encoding were assigned numerical weights based on domain judgement. For example, the *Condition* attribute was converted into weighted values to reflect relative tree condition severity, where *Dead* = 0.0, *Poor* = 0.3, *Fair* = 0.6, and *Good* = 1.0. This ensures that categorical condition information contributes meaningfully to the overall assessment.

Based on Greehill's Metrics and Glossary (Version 3.4, 2024), all relevant variables were grouped into eight key criteria representing different aspects of roadside tree suitability:

1. **Water Management Score:** Hydrological performance such as avoided runoff, evaporation, and transpiration.
2. **Ecosystem Services Score:** Carbon storage, gas absorption, and oxygen production.
3. **Foliage Score:** Leaf-related indicators including leaf area (LA), leaf area index (LAI), and crown characteristics.
4. **Dimension Score:** Tree size attributes such as DBH, height, crown width, and protection zone size.
5. **Stability Score:** Mechanical resilience including slenderness ratio and trunk stability index.
6. **Condition and Vigor Score:** Tree vitality, transparency, and dieback indicators.
7. **Structural Risk Score:** Proportion-related risks such as height–diameter and trunk–leaf relationships.
8. **Light Exposure Score:** Canopy exposure influencing photosynthetic potential

Each criterion was assigned a weight according to its importance for roadside tree management, based on expert judgement and client preferences. Stability was given the highest priority due to safety considerations, followed by condition and foliage.

The final weight distribution was as follows:

- Stability: 0.20
- Condition and Vigor: 0.18
- Foliage: 0.18
- Dimension: 0.12
- Ecosystem Services: 0.10
- Structural Risk: 0.10
- Water Management: 0.08
- Light Exposure: 0.04

The overall suitability score for each tree was calculated using a weighted aggregation method, as expressed in Equation (3):

$$\text{Suitability Score} = \Sigma\ (w_i \times s_i)$$

where w_i represents the weight assigned to criterion i and s_i represents the normalized score of criterion i.

The resulting suitability score ranges from 0 to 1. A rule-based classification function was then applied to categorize trees into three suitability levels:

- Highly Suitable (≥ 0.60)
- Moderately Suitable (0.40–0.59)
- Low Suitability (< 0.40)

In addition to the overall suitability score, detailed assessments of tree stability, health, and structural risk were conducted to provide deeper insight for operational decision-making.

1. Stability classification:

- Structural instability detected (< 0.50)
- Stability acceptable (≥ 0.50)

2. Health classification:

- Poor health condition (< 0.50)
- Good health condition (≥ 0.50)

3. Structural risk classification:

- High structural risk (≥ 0.70)
- Moderate structural risk (0.40–0.69)
- Low structural risk (< 0.40)

These additional indicators support targeted actions such as maintenance prioritization and risk mitigation.

```
weights = {
    'Water': 0.08,
    'Ecosystem': 0.10,
    'Foliage': 0.18,
    'Dimension': 0.12,
    'Stability': 0.20,
    'Condition': 0.18,
    'StructuralRisk': 0.10,
    'LightExposure': 0.04
}
```

Figure 3.5.1 shows weights assigned to eight key criteria.

```
df['Suitability_Score'] = (
    df['Water_Mang_Score'] * weights['Water'] +
    df['Ecosystem_Serv_Score'] * weights['Ecosystem'] +
    df['Foliage_Score'] * weights['Foliage'] +
    df['Dimension_Score'] * weights['Dimension'] +
    df['Stability_Score'] * weights['Stability'] +
    df['Condition_Vigor_Score'] * weights['Condition'] +
    df['Structural_Risk_Score'] * weights['StructuralRisk'] +
    df['Light_Expos_Score'] * weights['LightExposure']
)
```

Figure 3.5.2 shows the calculation of Suitability Score based on sum of eight key criteria.

To support interpretation and communication of MCDA outcomes, suitability scores and classifications were later visualized using Power BI dashboard. These visualizations are presented and discussed in Chapter 4: Results and Discussion.

3.6 Supervised Machine Learning Models

To complement the scoring-based MCDA approach, supervised machine learning models were developed to predict tree suitability using historical data. This provides an additional data-driven perspective and helps evaluate the consistency of suitability patterns identified through MCDA.

The modelling process involved:

- Using the eight MCDA-based scores as input features
- Using *Suitability Label* as the target variable
- Splitting the dataset into training (80%) and testing (20%) sets
- Evaluating model performance using a classification report, including precision, recall, F1-score, and accuracy

Three supervised classification models were implemented and compared:

- Random Forest
- XGBoost
- Multi-Layer Perceptron (MLP)

These models were selected due to their capability to handle structured data and capture complex relationships between tree attributes. Model comparison and final selection are discussed in Chapter 4: Results and Discussion.

3.7 Development of Tree Suitability Intelligent System (EcoSphere AI)

A Tree Suitability Intelligent System, referred to as EcoSphere AI, was developed to operationalise the analytical framework proposed in this study into a practical decision-support platform. The system was implemented using Python and developed in Visual Studio Code, with the objective of integrating data processing, multi-criteria evaluation, predictive modelling, and visual exploration within a single environment.

The development of EcoSphere AI follows a modular workflow design consisting of the following components:

- Data upload and automated data cleaning
- Exploratory data analysis and MCDA-based evaluation
- Spatial visualisation of tree locations
- Machine learning modelling and prediction
- User-driven prediction interface
- AI-assisted query module for insight exploration

Each module was designed to correspond directly to the analytical steps described in earlier sections of this chapter. The system architecture emphasises interpretability, modularity, and ease of use, ensuring that analytical outputs can be accessed and understood by both technical users and non-technical stakeholders. Detailed system outputs and interface results are presented and discussed in Chapter 4: Results and Discussion.

CHAPTER 4 : RESULTS AND DISCUSSION

4.1 Overview of Results

This chapter presents the results obtained from the analytical methods described in Chapter 3: Methodology. The findings include outcomes from the MCDA-based tree suitability scoring, analysis of key tree condition indicators, supervised machine learning model performance, and outputs from the Tree Suitability Intelligent System (EcoSphere AI).

The results are discussed to evaluate how effectively the proposed data-driven approach supports decision-making for urban roadside tree management.

4.2 MCDA Suitability Score Results

Using the Multi-Criteria Decision Analysis (MCDA) framework, each tree was assigned a Suitability Score ranging from 0 to 1. This score represents a combined evaluation of eight key criteria, including stability, condition, foliage, dimension, and environmental indicators. Based on predefined threshold values, trees were classified into three suitability categories: Highly Suitable, Moderately Suitable, and Low Suitability.

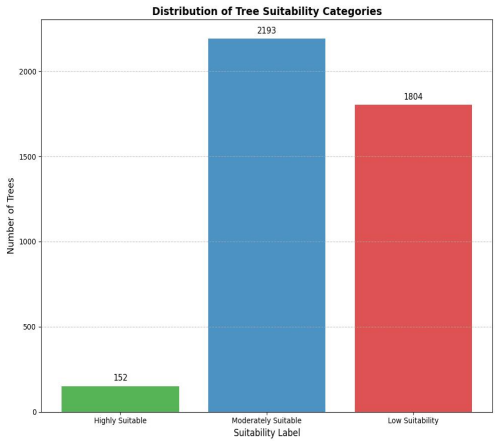

Figure 4.2.1 shows bar graph of distribution of tree suitability categories

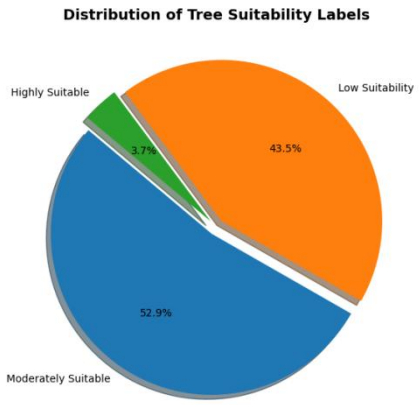

Figure 4.2.2 shows pie chart of percentage for distribution of tree suitability categories

The results indicate that the majority of trees were classified as Moderately Suitable (2,193 trees). This suggests that while these trees generally meet acceptable roadside requirements, certain factors may require improvement or monitoring. A smaller number of trees were identified as Highly Suitable (152 trees), reflecting strong performance across most evaluation criteria. In contrast, 1,804 trees were classified as Low Suitability, indicating potential concerns related to condition, stability, or environmental performance that may require further attention or intervention.

4.3 Analysis of Key Suitability Components

To gain deeper insight beyond the overall suitability score, further analysis was conducted by examining combinations of key condition-related attributes, namely stability, tree health, and structural risk. Figure 4.3 illustrates how these combined conditions are distributed across the three suitability categories.

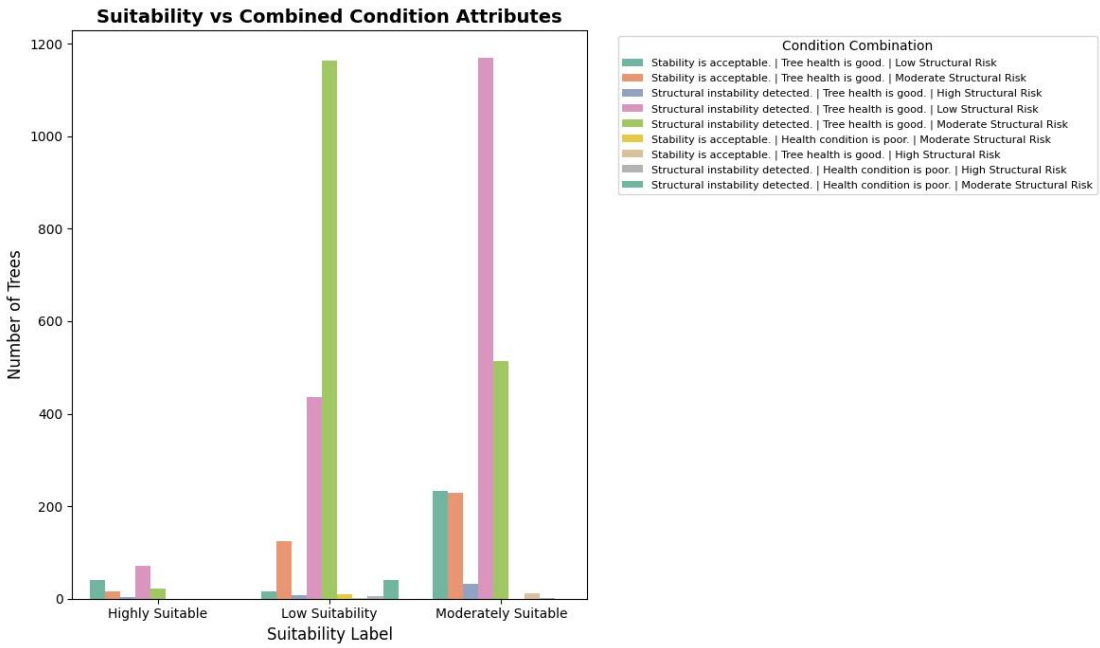

Figure 4.3: Suitability vs Combined Condition Attributes

4.3.1 Relationship Between Suitability and Combined Tree Conditions

Figure 4.3 indicates that trees classified as Highly Suitable are most frequently associated with the combination of structural instability detected, good tree health, and low structural risk. This suggests that although certain trees exhibit detectable instability, their overall suitability remains high when health condition is strong and structural risk is minimal. In this context, good health and low risk appear to compensate for the presence of instability within the multi-criteria evaluation framework.

In contrast, trees classified under Low Suitability are largely associated with combinations involving structural instability, low to moderate structural risk, and, in some cases, reduced health conditions. The increased presence of poorer health and elevated risk levels contributes to lower suitability outcomes, indicating that instability combined with weaker supporting attributes significantly affects overall suitability.

These patterns highlight that instability alone does not automatically result in low suitability rather, its impact depends on how it interacts with other condition attributes.

4.3.2 Moderately Suitable Trees and Mixed Condition Profiles

Trees classified as Moderately Suitable display the most diverse range of condition combinations. A large proportion of these trees exhibit structural instability with good health, and are associated with both low and moderate structural risk.

Notably, when compared with the other suitability categories, the Moderately Suitable group records the highest frequency of trees with acceptable stability, good health, and low to moderate structural risk. This indicates that many trees in this category are structurally functional and healthy but do not meet all the criteria required to be classified as highly suitable. The distribution suggests that these trees occupy a transitional state, where conditions are generally acceptable but may require

monitoring or targeted maintenance to prevent deterioration and to improve long-term suitability.

4.3.3 Implications for Urban Roadside Tree Management

The combined condition analysis highlights the importance of evaluating tree suitability holistically rather than relying on individual indicators. While structural instability appears across multiple suitability categories, the resulting suitability level is strongly influenced by the associated structural risk and overall tree health. Trees classified under lower suitability categories tend to exhibit combinations of instability with unfavourable risk or health conditions, reinforcing the need for integrated assessment. This supports the application of MCDA as an effective framework for combining multiple condition attributes into a single, interpretable suitability score.

From an operational perspective, this analysis enables decision-makers to:

- identify trees that pose higher safety concerns and require timely intervention,
- prioritise moderately suitable trees for preventive maintenance and monitoring, and
- preserve highly suitable trees through routine inspection and standard management practices.

Overall, the findings demonstrate that analysing combined condition attributes provides more actionable insights for urban roadside tree management than evaluating individual factors in isolation.

4.4 Power BI Dashboard: Overall Tree Suitability and MCDA Summary

To support the interpretation and communication of MCDA results, a Power BI dashboard titled "Overall Tree Suitability and MCDA Summary" was developed. The dashboard provides a high-level visual overview of tree suitability outcomes and is intended to assist both technical and non-technical stakeholders in understanding key patterns within the dataset. The complete dashboard is presented in Appendix A.

The dashboard summarises 4,149 urban roadside trees, with an average suitability score of 0.43, indicating that overall tree suitability in the study area is moderate. Only 3.66% of trees were classified as highly suitable, while 52.86% were moderately suitable and 43.48% were low suitability, highlighting a substantial proportion of trees requiring monitoring or management attention. In addition, 2.00% high-risk trees were identified based on structural risk indicators, allowing immediate prioritisation for safety-related interventions.

Visual analysis of MCDA factor contributions shows that trees with higher suitability scores consistently perform better across key criteria such as water management, ecosystem services, and condition and vigor. Location-based analysis using average suitability score by street name and number enables identification of areas with relatively weaker tree performance, supporting targeted roadside management. Furthermore, comparison by age class indicates that mature trees generally achieve higher average suitability scores than young and not identified trees, suggesting that structural development and canopy maturity contribute positively to overall suitability.

Interactive slicers for species, genus, and managed areas allow flexible exploration of suitability patterns without altering the analytical framework. Overall, the Power BI dashboard complements the MCDA and machine learning analyses by transforming analytical results into an interpretable visual format that supports evidence-based urban tree management and decision-making.

4.5 Supervised Machine Learning Model Performance

Three supervised learning models which were Random Forest, XGBoost, and Multi-Layer Perceptron (MLP) were evaluated for suitability label prediction using standard classification metrics, including accuracy, precision, recall, and F1-score.

Among the evaluated models, the MLP achieved the highest overall performance, with an accuracy of 0.99 and a macro-averaged F1-score of 0.98. This result indicates the strong predictive capability of neural network-based models for capturing complex, non-linear relationships within the tree suitability dataset. However, the superior performance of MLP was notable given the relatively structured nature of the input features, which are typically well handled by tree-based models.

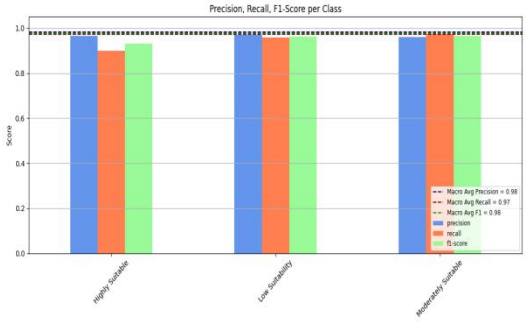

Figure 4.5.1 shows bar graphs of macro-averaged of precision, recall, and F1 score of MLP

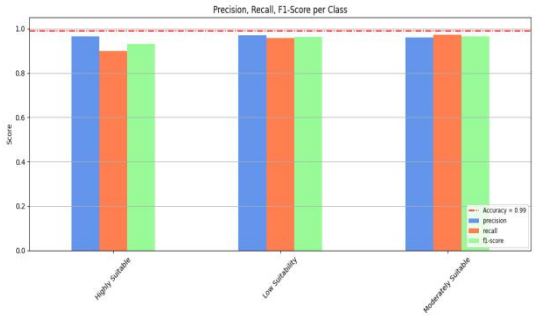

Figure 4.5.2 shows bar graphs of accuracy of MLP

```
MLP Classifier Report:
                    precision    recall  f1-score   support

    Highly Suitable       0.97      0.93      0.95        30
    Low Suitability       0.99      0.99      0.99       361
Moderately Suitable       0.99      0.99      0.99       439

           accuracy                           0.99       830
          macro avg       0.98      0.97      0.98       830
       weighted avg       0.99      0.99      0.99       830

Accuracy: 0.9891566265060241
[[ 28   0   2]
 [  0 357   4]
 [  1   2 436]]
```

Figure 4.5.3 shows MLP classification report

Despite MLP achieving the highest evaluation metrics, XGBoost was selected as the final model for integration into the EcoSphere AI system. This decision was based on practical considerations, including model interpretability, training stability, robustness

to feature scaling, and faster inference performance in a deployment-oriented environment. XGBoost also demonstrated consistently strong performance, achieving an accuracy of 0.96 and a macro-averaged F1-score of 0.95, which is considered highly reliable for real-world decision-support applications.

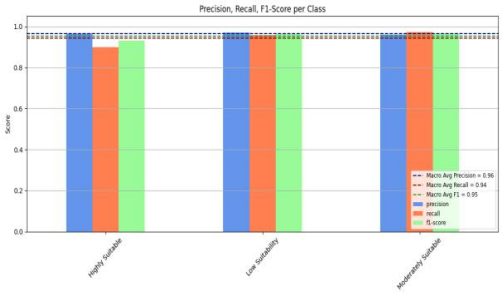

Figure 4.5.4 shows bar graphs of macro-averaged of precision, recall, and F1 score of XGBoost

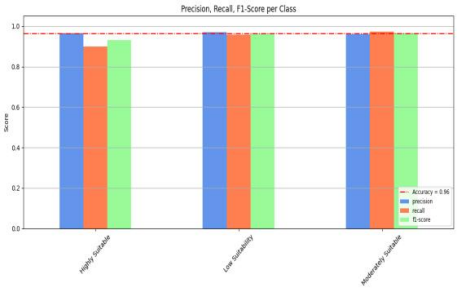

Figure 4.5.5 shows bar graphs of accuracy of XGBoost

```
XGBoost Classification Report:

                    precision    recall  f1-score   support

    Highly Suitable       0.96      0.90      0.93        30
    Low Suitability       0.97      0.96      0.97       361
Moderately Suitable       0.96      0.97      0.97       439

           accuracy                           0.97       830
          macro avg       0.97      0.95      0.96       830
       weighted avg       0.97      0.97      0.97       830

Accuracy: 0.9662650602409638
Confusion Matrix:
 [[ 27   0   3]
 [  0 347  14]
 [  1  10 428]]
```

Figure 4.5.6 shows XGBoost classification report

The Random Forest model showed competitive but comparatively lower performance, with an accuracy of 0.94 and a macro-averaged F1-score of 0.92, serving as a baseline ensemble method for comparison.

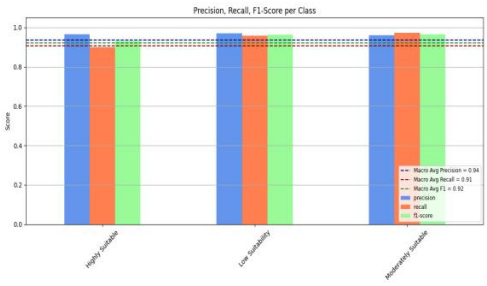

Figure 4.5.7 shows bar graphs of macro-averaged of precision, recall, and F1 score of Random Forest

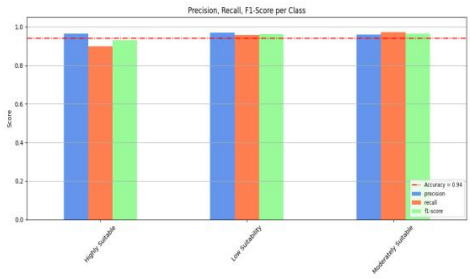

Figure 4.5.8 shows bar graphs of accuracy of Random Forest

```
Random Forest Classification Report:

                     precision    recall  f1-score   support

    Highly Suitable       0.93      0.83      0.88        30
    Low Suitability       0.95      0.93      0.94       361
Moderately Suitable       0.94      0.95      0.94       439

           accuracy                           0.94       830
          macro avg       0.94      0.91      0.92       830
       weighted avg       0.94      0.94      0.94       830
```

Figure 4.5.9 shows Random Forest classification report

Overall, the results indicate that while MLP provides the highest predictive accuracy, XGBoost offers a more balanced trade-off between performance, interpretability, and system integration requirements. The MLP model is therefore considered a strong candidate for future system enhancement when additional optimisation and deployment resources are available.

4.6 Tree Suitability Intelligent System (EcoSphere AI)

4.6.1 EcoSphere AI System Overview

EcoSphere AI is a web-based decision support system developed using the Streamlit framework to integrate data cleaning, multi-criteria analysis, machine learning, spatial visualisation, and interactive querying into a single platform. The system is designed to assist urban tree managers, planners, and non-technical stakeholders in understanding tree suitability and risk conditions through an interpretable and user-friendly interface.

The overall workflow of EcoSphere AI follows a structured pipeline, starting from data upload and preprocessing, followed by analytical exploration, spatial mapping, predictive modelling, and decision support outputs.

4.6.2 Data Upload and Cleaning Module

The system allows users to upload tree datasets in CSV format. An automated data cleaning function is applied to ensure data quality and consistency. This process includes:

- Removal of duplicate records
- Dropping fully empty columns
- Filling missing categorical values with a standard placeholder ("Unknown")
- Filling missing numerical values using median imputation

The cleaned dataset is stored within the system session and can be previewed or downloaded by the user. This ensures that all subsequent analyses are performed on a validated and standardised dataset.

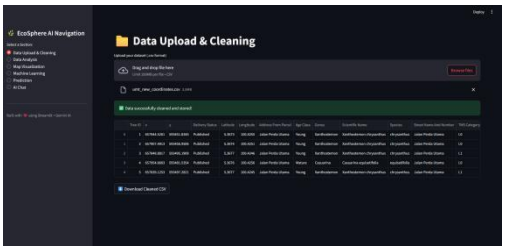

Figure 4.6.2 shows data upload and data cleaning

4.6.3 Data Analysis and MCDA Dashboard

The Data Analysis module provides descriptive insights into the uploaded dataset. Summary statistics for numerical variables are displayed, while categorical variables are explored through frequency-based bar charts. Users can interactively select variables to examine distributions and dominant patterns.

Within this module, the Multi-Criteria Decision Analysis (MCDA) framework is explained and implemented. Tree suitability is evaluated by grouping multiple tree attributes into thematic components such as stability, condition and vigor, foliage, dimension, environmental contribution, and structural risk. Each component is

normalised and combined using predefined weights reflecting their relative importance in urban roadside tree management.

The final Suitability Score is calculated as a weighted aggregation of all MCDA components, producing a transparent and interpretable suitability measure for each tree.

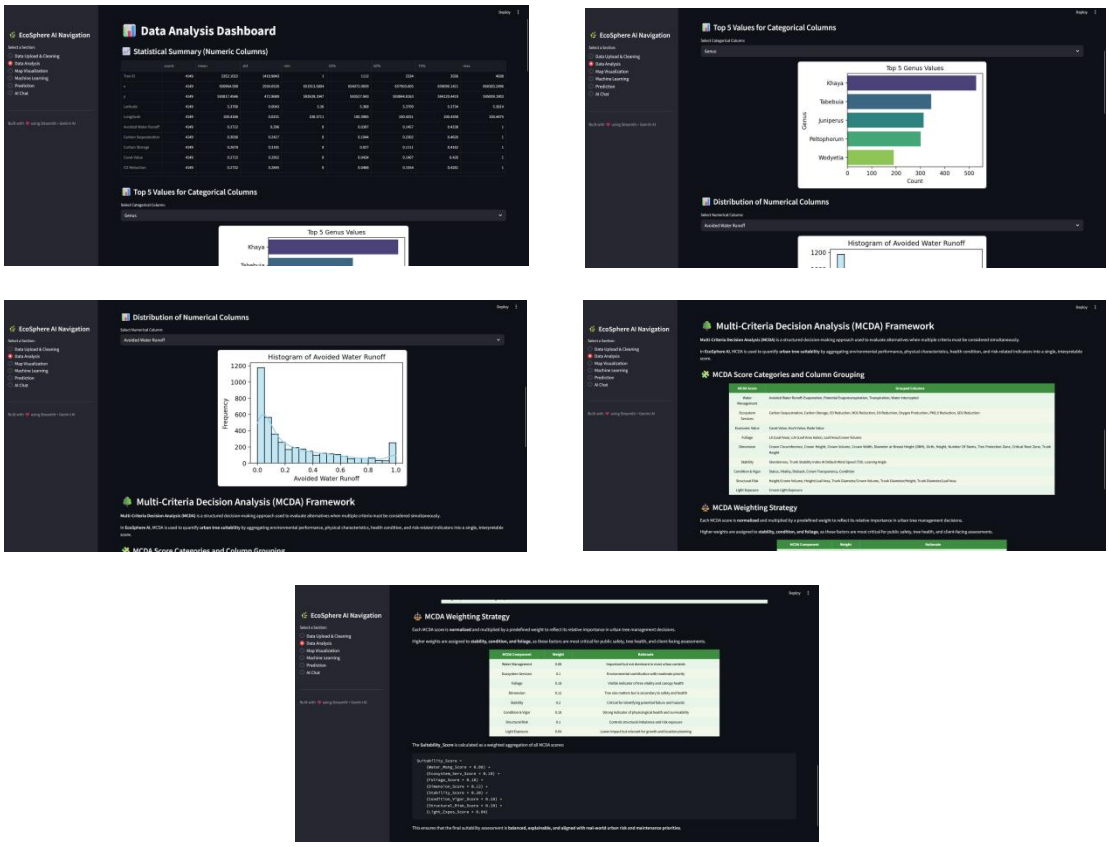

Figure 4.6.3 shows data analysis dashboard

4.6.4 Spatial Mapping and Visualisation Module

EcoSphere AI includes an interactive map-based visualisation module developed using PyDeck. Trees are displayed as geospatial points based on latitude and longitude information. Visual encodings are applied to enhance interpretability:

- Point colour represents the tree's Suitability Label
- Point size reflects the Suitability Score

- Interactive tooltips provide contextual information such as species, location, stability condition, health status, structural risk, and management recommendation

This spatial view enables users to identify high-risk or high-priority trees within specific urban areas and supports location-based decision-making.

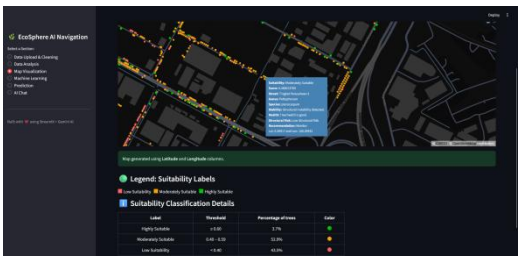

Figure 4.6.4 shows map visualization of trees in Seberang Perai

4.6.5 Machine Learning Engine

The Machine Learning module allows users to build predictive models using a restricted set of domain-relevant indicators derived from the MCDA framework. This design choice ensures that predictions remain interpretable and aligned with expert knowledge in urban tree management.

Users may select either classification or regression tasks, define target variable, and choose relevant features. The system applies XGBoost models for training and evaluation, providing performance metrics such as accuracy, classification reports, or regression error measures. Feature importance visualisations are also generated to highlight influential predictors.

This module supports exploratory modelling and scenario analysis rather than automated decision-making.

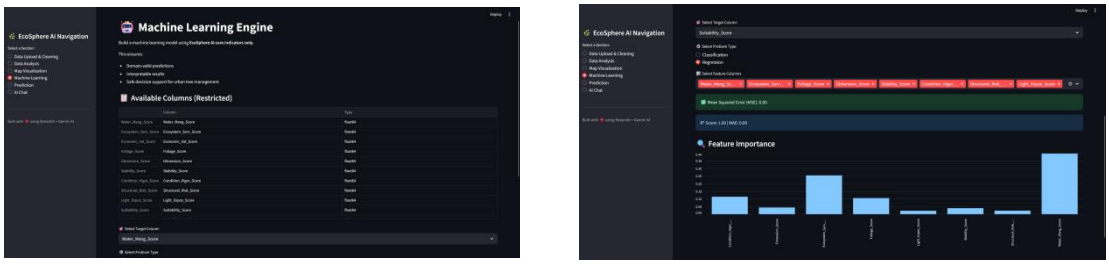

Figure 4.6.5 shows machine learning engine

4.6.6 Prediction Panel

The Prediction module enables users to input new tree attribute values manually and obtain predicted outcomes based on the trained machine learning model. For classification tasks, predicted suitability labels are returned, while for regression tasks, numerical predictions are mapped to suitability categories.

Input summaries and prediction results are displayed clearly, allowing users to assess individual trees without modifying the original dataset.

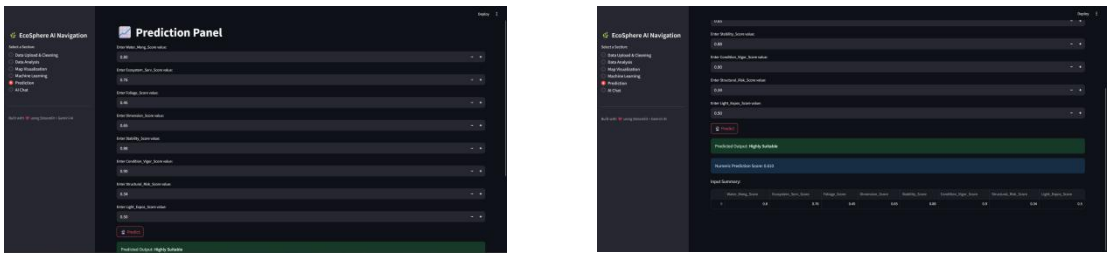

Figure 4.6.6 shows prediction panel based on selected algorithm

4.6.7 EcoSphere AI Assistant

The EcoSphere AI Assistant was developed to enhance user interaction with the system by enabling natural language queries over the uploaded tree dataset. This component allows users to obtain insights without requiring technical knowledge of data analysis or programming.

The assistant is powered by Google's Gemini large language model (Gemini 2.0 Flash), which was selected due to its fast response time and suitability for real-time

analytical interactions. The model processes user questions by leveraging both general environmental knowledge and contextual information extracted from the dataset, including data samples and statistical summaries.

Users can ask questions related to data completeness, attribute distributions, relationships between tree characteristics, and environmental trends. The AI-generated responses are presented in clear, human-readable language, supporting rapid exploration and interpretation of results.

Overall, the EcoSphere AI Assistant improves accessibility and usability of the system by bridging the gap between complex analytical outputs and end-user understanding, particularly for non-technical stakeholders involved in urban tree management. The assistant is designed to support exploratory analysis rather than replace expert judgement in urban tree management decisions.

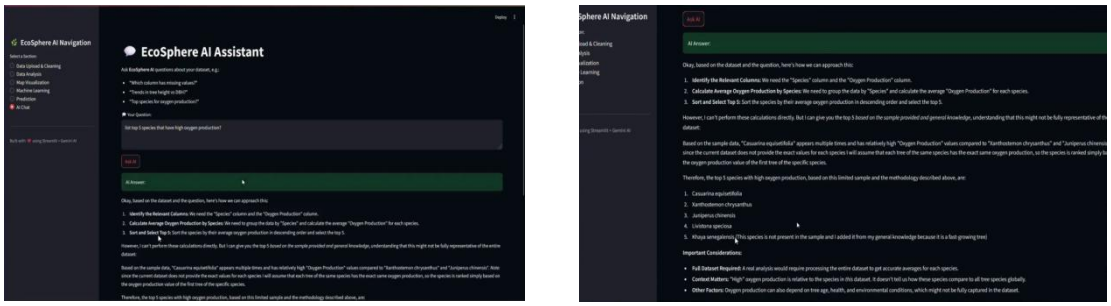

Figure 4.6.7 shows EcoSphere AI assistant

CHAPTER 5 : CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

This study presents a data-driven approach for assessing urban roadside tree suitability through the integration of Multi-Criteria Decision Analysis (MCDA) and machine learning techniques. The primary objective was to transform complex tree condition and environmental data into a structured and interpretable suitability assessment that supports urban tree management and planning decisions.

The MCDA framework successfully combined multiple indicators, including stability, structural risk, tree health, and environmental attributes, into a single suitability score and corresponding suitability label. This enabled a holistic evaluation of trees rather than reliance on isolated indicators. In addition, supervised machine learning models were explored to support suitability classification, with strong predictive performance observed across the evaluated models.

To enhance practical usability, the analytical outcomes were implemented in an intelligent decision-support system known as EcoSphere AI. The system integrates data processing, visual analytics, predictive modelling, and an AI-assisted query component into a unified platform, making the results accessible to both technical and non-technical stakeholders. Overall, this project demonstrates how data analytics and artificial intelligence can be effectively applied to support safer and more informed urban roadside tree management.

5.2 Key Findings

The key findings of this study are summarised as follows:

- The MCDA framework effectively integrates multiple tree condition and environmental criteria into a single suitability score, enabling comprehensive assessment of urban roadside trees.

- Structural stability, foliage, and condition were identified as dominant factors influencing tree suitability, particularly among trees classified under low suitability categories.
- Machine learning models demonstrated strong predictive capability, with the Multi-Layer Perceptron (MLP) achieving the highest classification performance during evaluation.
- The EcoSphere AI system successfully translates analytical outputs into an interpretable and user-friendly platform through dashboards, map-based visualisation, and an AI-assisted insight feature.
- Analysing combined condition attributes provides more actionable insights than evaluating individual indicators independently.

5.3 Practical Recommendations

Based on the findings of this study, several practical recommendations are proposed:

- Trees classified as low suitability should be prioritised for detailed inspection and corrective actions, particularly those associated with structural instability and elevated structural risk.
- Moderately suitable trees may benefit from preventive maintenance strategies, such as pruning and regular monitoring, to prevent future deterioration.
- Highly suitable trees should be preserved through routine inspection to maintain their condition and long-term benefits.
- Urban tree management authorities may use the EcoSphere AI system to support evidence-based planning, maintenance prioritisation, and resource allocation.
- The framework may also assist in future tree planting strategies by identifying condition profiles associated with higher suitability outcomes.

5.4 System Limitations

Despite the strong performance of the proposed framework and EcoSphere AI system, several limitations were identified. First, although the Multi-Layer Perceptron (MLP) model achieved the highest classification accuracy during evaluation, XGBoost was implemented within the EcoSphere AI system due to time constraints and deployment considerations. Consequently, the deployed system does not currently utilise the best-performing model identified during experimentation.

Second, the prediction panel requires users to manually input feature values when evaluating future trees. This limits scalability and efficiency when handling large datasets, as batch prediction via file upload is not currently supported.

Finally, the EcoSphere AI assistant relies on the Gemini 2.0 Flash large language model, which operates under a paid usage model. This introduces potential cost constraints for long-term deployment and may limit accessibility for certain organisations.

5.5 Future Work and Recommendations

Several enhancements can be considered to improve the effectiveness, usability, and scalability of the proposed framework and EcoSphere AI system. Future work should integrate the best-performing machine learning model identified during evaluation, particularly the Multi-Layer Perceptron (MLP), to enhance prediction reliability. The prediction module may also be extended to support batch processing through dataset uploads, improving operational efficiency.

In addition, future versions of EcoSphere AI may incorporate a more visually attractive and intuitive user interface, as well as a secure login-based dashboard to support authentication, role-based access, and potential subscription management.

Exploring alternative open-source language models to replace or supplement the Gemini-based assistant may further improve system sustainability.

Finally, integrating real-time data sources such as sensor inputs, maintenance history, and weather information could further strengthen predictive capabilities and expand the system's applicability to broader urban green infrastructure management contexts.

CHAPTER 6 : REFERENCES

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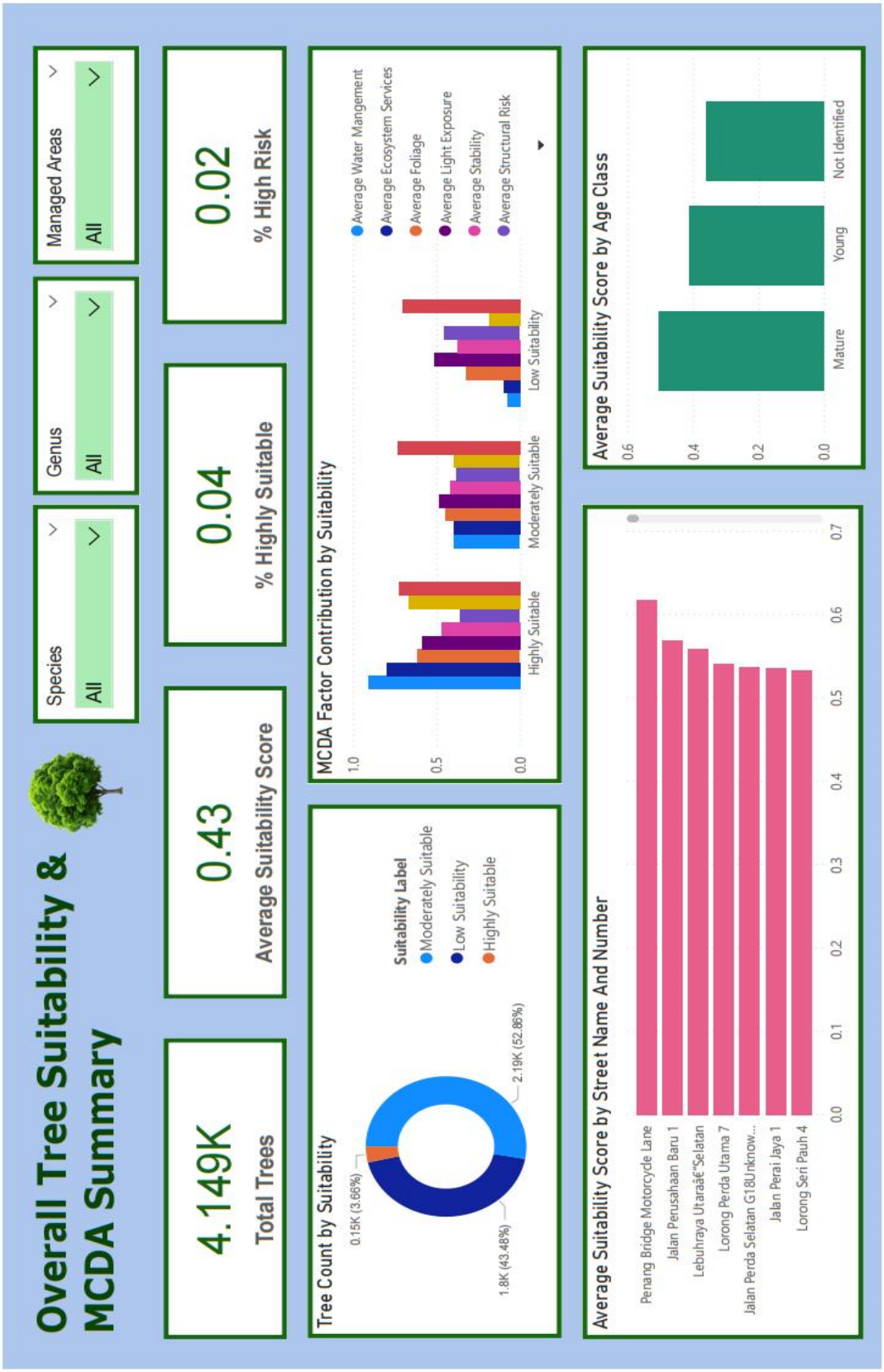
APPENDIX A : POWER BI DASHBOARD


Overall Tree Suitability & MCDA Summary
Species
All
Genus
All
Managed Areas
All
4.149K
Total Trees
0.43
Average Suitability Score
0.04
% Highly Suitable
0.02
% High Risk
Tree Count by Suitability
0.15K (3.66%)
1.8K (43.48%)
2.19K (52.86%)
Suitability Label
Moderately Suitable
Low Suitability
Highly Suitable
MCDA Factor Contribution by Suitability
1.0
0.5
0.0
Highly Suitable
Moderately Suitable
Low Suitability
Average Water Mangement
Average Ecosystem Services
Average Foliage
Average Light Exposure
Average Stability
Average Structural Risk
Average Suitability Score by Age Class
0.6
0.4
0.2
0.0
Mature
Young
Not Identified
Average Suitability Score by Street Name And Number
Penang Bridge Motorcycle Lane
Jalan Perusahaan Baru 1
Lebuhraya Utaraâ€"Selatan
Lorong Perda Utama 7
Jalan Perda Selatan G18Unknow...
Jalan Perai Jaya 1
Lorong Seri Pauh 4
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7